COMPLEX STABLE PHASE RETRIEVAL WITH ONE EXCEPTIONAL COORDINATE

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Abstract. We give a compactness proof of the complex one-exception stable phase retrieval theorem. The proof is organized in the same head-tail architecture as the real one-exception argument: one first proves the coefficient decompositions and the head identification, then extracts a limit model in which a head pair and a tail pair are independent and satisfy an exact modulus identity, and finally proves a support theorem for the tail difference. Two complementary support routes are recorded for the degenerate head case. The first is purely geometric: cone-supported infinitely divisible laws are supported on additive subgroups of the cone, hence on either a complex line or a conjugate plane, and both possibilities force the tail difference onto a real line. The second, included as a remark after the proof, uses the nontriviality of the Gaussian part of a diffuse tail limit. Along the way we isolate the quantitative angular-spread estimate needed for the complex cross-coordinate probes and we import the rigorously proved compactness ingredients from the general complex stable phase retrieval theorem.

1. Introduction

Let (Ω, μ) be a probability space and let $\xi_1, \dots, \xi_n$ be independent complex-valued random variables. The main theorem of [1] states that if every ξ_j satisfies a two-sided L^1-L^2 norm-equivalence bound and a uniform lower bound on the generalized-circle gap $\mathrm{gap}_{\mathrm{gc}}(\xi_j) \geq \kappa > 0$, then there is a uniform orthogonal modulus gap

$$\big\| \, |X_a| - |X_b| \, \big\|_{L^2} \geq c(A, B, \kappa) > 0$$

for every pair of orthogonal unit vectors $a, b \in \mathbb{C}^n$.

The theorem proved here is the one-exception analogue of [1, Theorem 2.1]: a single exceptional coordinate η is allowed to violate the lower L^1-L^2 bound and the generalized-circle gap, while the good coordinates retain the same uniform hypotheses as in the fully homogeneous theorem.

The present proof keeps the current `.tex` file as the main source but merges into it the rigorously proved compactness ingredients from Appendix A of [2]. The resulting organization mirrors the real one-exception proof from [3]. First one proves the head-tail decompositions and the head identification by bounded probes. Next one packages those lemmas into a compactness reduction, whose output is an exact modulus identity

$$|H_X + R_X| = |H_Y + R_Y|$$

for an independent head pair and tail pair. The proof then culminates in a support theorem for the tail difference. When the head is nontrivial, the support theorem follows from a geometric elimination of the head variable. In the degenerate case, one uses the additive-subgroup structure of cone-supported infinitely divisible laws to force the tail difference onto a real line.

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For completeness, we also keep the Gaussian route to the degenerate case. That route is not needed in the final proof, but it gives an independent way to see that the same support conclusion follows from the nontriviality of the Gaussian part of a diffuse tail limit.

2. Background and notation

Definition 1 (Generalized circle)**.** *A* generalized circle *in* $\mathbb{C}$ *is either a Euclidean circle* $\{z \in \mathbb{C} : |z - c| = r\}$ *with* $c \in \mathbb{C}$ *and* $r \geq 0$*, or a real affine line in* $\mathbb{C} \cong \mathbb{R}^2$*.*

Definition 2 (Generalized-circle gap [1, Definitions 3.3–3.4])**.** *For a complex-valued random variable* $\xi \in L^2(\mu)$*, define*

$$\mathrm{gap}_{\mathrm{gc}}(\xi) := \inf_{\substack{A,C\in\mathbb{R},\ B\in\mathbb{C} \\ A^2+2|B|^2+C^2=1}} \left\| A|\xi|^2 + 2\,\Re(B\xi) + C \right\|_{L^1}.$$

By [1, Lemma 3.8], $\mathrm{gap}_{\mathrm{gc}}(\xi) = 0$ *if and only if* $\mathrm{supp}(\xi)$ *is contained in a generalized circle.*

Definition 3 (Phase-invariant distance [1, Definition 2.2])**.** *For* $f, g \in L^2(\mu)$*, set*

$$d_{\mathbb{T}}(f, g) := \inf_{\lambda\in\mathbb{T}} \|f - \lambda g\|_{L^2(\mu)}.$$

We use the following lemmas from [1].

Lemma 4 (One-coordinate rigidity [1, Lemma 3.10])**.** *Let* $S \subset \mathbb{C}$ *be a set not contained in any generalized circle. Let* $\alpha, \beta, u, v \in \mathbb{C}$ *and assume*

$$|\alpha t + u| = |\beta t + v| \qquad \text{for every } t \in S.$$

Then $|\alpha| = |\beta|$ *and* $\bar{\alpha}\, u = \bar{\beta}\, v$*. Consequently, either* $\alpha = \beta = 0$*, or there exists* $\lambda \in \mathbb{T}$ *such that* $\beta = \lambda\alpha$ *and* $v = \lambda u$*.*

Lemma 5 (Support of an independent pair [1, Lemma 3.11])**.** *If* U *and* V *are independent random variables taking values in Polish spaces, then* $\mathrm{supp}(U, V) = \mathrm{supp}(U) \times \mathrm{supp}(V)$*.*

Lemma 6 (Almost-sure vanishing extends to the support [1, Lemma 3.12])**.** *Let* S *be a Polish space, let* Z *be an* S*-valued random variable, and let* $F : S \to \mathbb{R}$ *be continuous. If* $F(Z) = 0$ *almost surely, then* $F(x) = 0$ *for every* $x \in \mathrm{supp}(Z)$*.*

3. Main result and contradiction setup

Theorem 7 (Complex one-exception orthogonal lower bound)**.** *Let* η *be a complex-valued random variable satisfying*

$$\mathbb{E}[\eta] = 0, \qquad \mathbb{E}[|\eta|^2] = 1.$$

Fix constants $0 < A \leq B < 1$ *and* $\kappa > 0$*. Then there exists*

$$c = c(A, B, \kappa, \eta) > 0$$

such that the following holds for every $n \in \mathbb{N}$*.*

Let $\xi_1, \ldots, \xi_n$ *be complex-valued random variables, independent of each other and of* η*, satisfying*

$$\mathbb{E}[\xi_j] = 0, \qquad \mathbb{E}[|\xi_j|^2] = 1 \qquad (1 \leq j \leq n),$$

and assume:

(1) *for every* $u = (u_1, \ldots, u_n) \in \mathbb{C}^n$*,*

$$A\|u\|_2 \leq \Big\| \sum_{j=1}^{n} u_j \xi_j \Big\|_{L^1} \leq B\|u\|_2;$$

(2) $\mathrm{gap}_{\mathrm{gc}}(\xi_j) \geq \kappa$ *for every* $1 \leq j \leq n$*.*

For every $a = (a_0, a_1, \dots, a_n)$ *and* $b = (b_0, b_1, \dots, b_n)$ *in* $\mathbb{C}^{n+1}$ *with*

$$\|a\|_2 = \|b\|_2 = 1, \qquad \langle a, b\rangle = 0,$$

define

$$X_a := a_0\eta + \sum_{j=1}^{n} a_j\xi_j, \qquad X_b := b_0\eta + \sum_{j=1}^{n} b_j\xi_j.$$

Then

$$\big\| \, |X_a| - |X_b| \, \big\|_{L^2} \geq c.$$

Remark 8. *Theorem 7 is the natural one-exception analogue of* [1, Theorem 2.1]. *The constant is allowed to depend on the full law of the exceptional variable* η.

Corollary 9 (Complex one-exception stable phase retrieval)**.** *Under the hypotheses of Theorem 7, let* $E_n := \operatorname{span}\{\eta, \xi_1, \dots, \xi_n\} \subset L^2(\mu)$. *There exists* $C = C(A, B, \kappa, \eta) < \infty$ *such that for all* $f, g \in E_n$,

$$d_{\mathbb{T}}(f, g) \leq C \, \big\| \, |f| - |g| \, \big\|_{L^2(\mu)}.$$

Proof. This is proved exactly as in [1, Corollary 2.4] once the orthogonal lower bound is known. □

Proposition 10 (Normalized counterexample array)**.** *If Theorem 7 fails, then there exist:*

(i) *a triangular array* $(\xi_{n,i})_{n\geq 1,\ i\geq 1}$ *such that for each fixed* n *the family*

$$(\eta, \xi_{n,1}, \xi_{n,2}, \dots)$$

is independent, every $\xi_{n,i}$ *is centered and satisfies*

$$\mathbb{E}[|\xi_{n,i}|^2] = 1, \qquad \operatorname{gap}_{\mathrm{gc}}(\xi_{n,i}) \geq \kappa,$$

and for every finitely supported $u \in \ell^2(\mathbb{N})$,

$$A\|u\|_2 \leq \Big\| \sum_{i\geq 1} u_i \xi_{n,i} \Big\|_{L^1} \leq B\|u\|_2;$$

(ii) *vectors* $a^{(n)}, b^{(n)} \in \ell^2(\{0, 1, 2, \dots\})$ *such that*

$$\|a^{(n)}\|_2 = \|b^{(n)}\|_2 = 1, \qquad \langle a^{(n)}, b^{(n)}\rangle = 0,$$

and, writing

$$X_n := a_0^{(n)}\eta + \sum_{i\geq 1} a_i^{(n)}\xi_{n,i}, \qquad Y_n := b_0^{(n)}\eta + \sum_{i\geq 1} b_i^{(n)}\xi_{n,i},$$

one has

$$\big\| \, |X_n| - |Y_n| \, \big\|_{L^2} \longrightarrow 0.$$

Proof. This is the standard contradiction setup: one takes violating orthogonal unit pairs row by row, and pads finite vectors by zeros. □

4. Coefficient decompositions and the limit model

We now formulate the compactness scheme in the same language as [3]. The point is that the proof should work directly with the bad sequence of sums rather than first extracting limit laws for the individual coordinates.

Lemma 11 (Reordered profile decomposition)**.** *After passing to a subsequence in Proposition 10 and, in each row, applying the same permutation to the good coordinates and to their coefficients, one may find vectors*

$$a, b \in \ell^2(\{0, 1, 2, \dots\})$$

and integers $m_n \to \infty$ such that, with

$$a^{(n)}_{\text{head}} := a^{(n)} \mathbf{1}_{\{0,1,\dots,m_n\}}, \qquad a^{(n)}_{\text{tail}} := a^{(n)} \mathbf{1}_{\{m_n+1, m_n+2, \dots\}},$$

and likewise for $b^{(n)}$, the following hold:

(1) $a^{(n)}_{\text{head}} \to a$ *and* $b^{(n)}_{\text{head}} \to b$ *strongly in* ℓ^2*;*

(2)
$$\max_{i > m_n} \max\big(|a^{(n)}_i|, |b^{(n)}_i|\big) \longrightarrow 0;$$

(3)
$$\|a^{(n)}_{\text{tail}}\|_2^2 \longrightarrow 1 - \|a\|_2^2, \qquad \|b^{(n)}_{\text{tail}}\|_2^2 \longrightarrow 1 - \|b\|_2^2,$$

and

$$\langle a^{(n)}_{\text{tail}}, b^{(n)}_{\text{tail}} \rangle \longrightarrow -\langle a, b \rangle.$$

Proof. The proof is the same monotone-reordering and diagonal extraction used in the real one-exception paper; it depends only on the coefficient vectors. □

Remark 12. *The exceptional coordinate η is kept fixed at index 0. Only the good coordinates are reordered.*

Proposition 13 (Bounded probes for good coordinates)**.** *Let ξ be a centered complex-valued random variable satisfying*

$$\mathbb{E}[|\xi|^2] = 1, \qquad A \le \|\xi\|_{L^1} \le B < 1.$$

Then there exist bounded measurable functions $S, T : \Omega \to \mathbb{C}$ such that

$$\mathbb{E}[S] = \mathbb{E}[T] = 0, \qquad \mathbb{E}[\xi \overline{S}] = \mathbb{E}[\overline{\xi} S] = 1, \qquad \mathbb{E}[(|\xi|^2 - 1)T] = 1,$$

and

$$\|S\|_{L^\infty} \le \frac{2}{A}, \qquad \|T\|_{L^\infty} \le \frac{2}{1 - B}.$$

Proof. Define the phase function

$$\phi := \begin{cases} \xi/|\xi|, & \xi \neq 0, \\ 0, & \xi = 0. \end{cases}$$

Then $|\phi| \le 1$ and

$$\mathbb{E}[\xi \overline{\phi}] = \mathbb{E}[|\xi|] = \|\xi\|_{L^1} \ge A.$$

Set

$$S := \frac{\phi - \mathbb{E}[\phi]}{\mathbb{E}[|\xi|]}.$$

Since $\mathbb{E}[\xi] = 0$, one has

$$\mathbb{E}[\xi \overline{S}] = \frac{\mathbb{E}[\xi \overline{\phi}] - \mathbb{E}[\xi]\mathbb{E}[\overline{\phi}]}{\mathbb{E}[|\xi|]} = 1.$$

Taking complex conjugates gives $\mathbb{E}[\overline{\xi} S] = 1$, and clearly $\mathbb{E}[S] = 0$. The bound

$$\|S\|_{L^\infty} \le \frac{2}{\mathbb{E}[|\xi|]} \le \frac{2}{A}$$

is immediate.

Next let

$$\psi := \operatorname{sign}(|\xi|^2 - 1),$$

which is real-valued and bounded by 1 in modulus. Then

$$\mathbb{E}[(|\xi|^2-1)\psi] = \mathbb{E}\big[\,||\xi|^2-1|\,\big].$$

Moreover,

$$||\xi|^2-1| = ||\xi|-1|(|\xi|+1) \geq ||\xi|-1|,$$

so

$$\mathbb{E}\big[\,||\xi|^2-1|\,\big] \geq \mathbb{E}[\,||\xi|-1|\,] \geq 1-\mathbb{E}[|\xi|] \geq 1-B.$$

Therefore, with

$$T := \frac{\psi-\mathbb{E}[\psi]}{\mathbb{E}[(|\xi|^2-1)\psi]},$$

we get $\mathbb{E}[T]=0$,

$$\mathbb{E}[(|\xi|^2-1)T] = 1,$$

and

$$\|T\|_{L^\infty} \leq \frac{2}{\mathbb{E}[\,||\xi|^2-1|\,]} \leq \frac{2}{1-B}.$$

□

Lemma 14 (Quantitative angular spread)**.** *Let ξ be a centered complex-valued random variable with*

$$\mathbb{E}[|\xi|^2] = 1.$$

Define

$$m := \mathbb{E}[|\xi|], \qquad c := \mathbb{E}\left[\frac{\xi^2}{|\xi|}\right],$$

where $\xi^2/|\xi|$ is interpreted as 0 on $\{\xi=0\}$. Then

$$|c| \leq m - \mathrm{gap}_{\mathrm{gc}}(\xi)^2.$$

In particular, if S is the linear probe from Proposition 13, then

$$\rho := \mathbb{E}[\xi S] = \frac{c}{m}$$

satisfies

$$|\rho| \leq 1-\mathrm{gap}_{\mathrm{gc}}(\xi)^2 \leq 1.$$

Proof. If $c=0$, there is nothing to prove. Otherwise choose $\omega\in\mathbb{T}$ so that

$$\omega^2 c = |c|.$$

Write $\omega\xi = |\xi|e^{it}$. Then

$$|c| = \Re(\omega^2 c) = \mathbb{E}[|\xi|\cos(2t)]\,,$$

so

$$m-|c| = \mathbb{E}\big[|\xi|\big(1-\cos(2t)\big)\big] = 2\,\mathbb{E}\big[|\xi|\sin^2 t\big]\,.$$

Consider the generalized circle given by the line

$$\Im(\omega z) = 0.$$

The corresponding normalized test has the form

$$z \longmapsto \sqrt{2}\,\Im(\omega z),$$

and therefore

$$\mathrm{gap}_{\mathrm{gc}}(\xi) \leq \sqrt{2}\,\mathbb{E}[|\Im(\omega\xi)|] = \sqrt{2}\,\mathbb{E}[|\xi||\sin t|]\,.$$

By Cauchy–Schwarz,

$$\mathbb{E}[|\xi||\sin t|]^2 \leq \mathbb{E}[|\xi|]\,\mathbb{E}\big[|\xi|\sin^2 t\big] = \frac{m(m-|c|)}{2}.$$

Combining the last two displays gives

$$\mathrm{gap}_{\mathrm{gc}}(\xi)^2 \le m(m-|c|) \le m-|c|,$$

because $m \le \|\xi\|_{L^2} = 1$. This proves the first claim. The identity

$$\mathbb{E}[\xi S] = \frac{\mathbb{E}[\xi^2/|\xi|]}{\mathbb{E}[|\xi|]} = \frac{c}{m}$$

follows directly from the definition of S, and the stated bound on $|\rho|$ is immediate. □

Proposition 15 (Linear probe for the exceptional coordinate)**.** *Let η be a centered complex-valued random variable satisfying*

$$\mathbb{E}[|\eta|^2] = 1.$$

Then there exists a bounded measurable function S_η such that

$$\mathbb{E}[S_\eta] = 0, \qquad \mathbb{E}[\eta\overline{S_\eta}] = \mathbb{E}[\overline{\eta}S_\eta] = 1.$$

Proof. Define

$$\phi_\eta := \begin{cases} \eta/|\eta|, & \eta \ne 0, \\ 0, & \eta = 0. \end{cases}$$

Then $\mathbb{E}[|\eta|] > 0$ because $\mathbb{E}[|\eta|^2] = 1$. Set

$$S_\eta := \frac{\phi_\eta - \mathbb{E}[\phi_\eta]}{\mathbb{E}[|\eta|]}.$$

Exactly as above, $\mathbb{E}[S_\eta] = 0$ and

$$\mathbb{E}[\eta\overline{S_\eta}] = \mathbb{E}[\overline{\eta}S_\eta] = 1.$$

□

Proposition 16 (Head identification)**.** *Assume the setup of Lemma 11. Then either:*

(1) *there exists $\tau \in \mathbb{T}$ such that*

$$a = \tau b$$

and

$$\|a^{(n)}_{\mathrm{head}} - \tau b^{(n)}_{\mathrm{head}}\|_2 \longrightarrow 0;$$

(2) *or else*

$$a_i = b_i = 0 \qquad \text{for every } i \ge 1.$$

Proof. Since

$$\big\||X_n|^2 - |Y_n|^2\big\|_{L^1} \le \big\||X_n| - |Y_n|\big\|_{L^2}\big\||X_n| + |Y_n|\big\|_{L^2} \le 2\big\||X_n| - |Y_n|\big\|_{L^2},$$

we have

$$\big\||X_n|^2 - |Y_n|^2\big\|_{L^1} \longrightarrow 0.$$

Fix $i \ge 1$. Let $T_{n,i}$ be the quadratic probe given by Proposition 13 for $\xi_{n,i}$. Since $T_{n,i}$ is a bounded centered function of $\xi_{n,i}$ alone, it is independent of η and of all $\xi_{n,j}$ with $j \ne i$. Hence

$$\mathbb{E}\big[(|X_n|^2 - |Y_n|^2)T_{n,i}\big] \longrightarrow 0.$$

Expanding $|X_n|^2 = X_n\overline{X_n}$ and using independence and centering, all terms vanish except the one involving $(|\xi_{n,i}|^2 - 1)T_{n,i}$. Therefore

$$\mathbb{E}\big[(|X_n|^2 - |Y_n|^2)T_{n,i}\big] = |a_i^{(n)}|^2 - |b_i^{(n)}|^2.$$

Passing to the limit gives

$$|a_i|^2 = |b_i|^2 \qquad (i \ge 1).$$

Now fix distinct indices $i, j \geq 1$. Let $S_{n,i}$ and $S_{n,j}$ be the linear probes from Proposition 13. Since $\overline{S_{n,i}} S_{n,j}$ is bounded and depends only on $(\xi_{n,i}, \xi_{n,j})$, we again have

$$\mathbb{E}\big[(|X_n|^2 - |Y_n|^2)\overline{S_{n,i}} S_{n,j}\big] \longrightarrow 0.$$

Set

$$A_{n,i,j} := a_i^{(n)} \overline{a_j^{(n)}} - b_i^{(n)} \overline{b_j^{(n)}}, \qquad \rho_{n,\ell} := \mathbb{E}[\xi_{n,\ell} S_{n,\ell}] \qquad (\ell = i, j).$$

As before, independence and centering eliminate every term except the two containing $\xi_{n,i}\overline{\xi_{n,j}}$ and $\xi_{n,j}\overline{\xi_{n,i}}$. Using

$$\mathbb{E}[\xi_{n,i}\overline{S_{n,i}}] = 1, \qquad \mathbb{E}[\overline{\xi_{n,j}} S_{n,j}] = 1,$$

and taking complex conjugates where needed, we obtain

$$\mathbb{E}\big[(|X_n|^2 - |Y_n|^2)\overline{S_{n,i}} S_{n,j}\big] = A_{n,i,j} + \overline{A_{n,i,j}}\, \rho_{n,j}\, \overline{\rho_{n,i}}.$$

By Lemma 14 and the standing lower bound $\mathrm{gap}_{\mathrm{gc}}(\xi_{n,\ell}) \geq \kappa$, one has

$$|\rho_{n,\ell}| \leq 1 - \kappa^2 \qquad (\ell = i, j).$$

Hence

$$\big|A_{n,i,j}\big| \leq \frac{\big|\mathbb{E}\big[(|X_n|^2 - |Y_n|^2)\overline{S_{n,i}} S_{n,j}\big]\big|}{1 - (1-\kappa^2)^2} \longrightarrow 0,$$

because

$$\big|A_{n,i,j} + \overline{A_{n,i,j}}\, \rho_{n,j}\, \overline{\rho_{n,i}}\big| \geq \big(1 - |\rho_{n,j}\rho_{n,i}|\big)\, |A_{n,i,j}|.$$

Passing to the limit yields

$$a_i \overline{a_j} = b_i \overline{b_j} \qquad (i \neq j,\ i, j \geq 1).$$

Let S_η be given by Proposition 15. For every $j \geq 1$, letting $S_{n,j}$ be the good-coordinate linear probe for $\xi_{n,j}$, we likewise obtain

$$\mathbb{E}\big[(|X_n|^2 - |Y_n|^2)\overline{S_\eta} S_{n,j}\big] \longrightarrow 0$$

and therefore

$$a_0^{(n)} \overline{a_j^{(n)}} - b_0^{(n)} \overline{b_j^{(n)}} \longrightarrow 0.$$

Passing to the limit gives

$$a_0 \overline{a_j} = b_0 \overline{b_j} \qquad (j \geq 1).$$

Assume first that there exists $i_0 \geq 1$ with $a_{i_0} \neq 0$. Since $|a_{i_0}| = |b_{i_0}|$, there exists $\tau \in \mathbb{T}$ such that

$$a_{i_0} = \tau b_{i_0}.$$

For every $j \geq 1$,

$$a_{i_0} \overline{a_j} = b_{i_0} \overline{b_j} = \overline{\tau} a_{i_0} \overline{b_j},$$

so $a_j = \tau b_j$. Taking $j = i_0$ in the exceptional-good identity gives

$$a_0 \overline{a_{i_0}} = b_0 \overline{b_{i_0}} = \tau b_0 \overline{a_{i_0}},$$

and therefore $a_0 = \tau b_0$. Thus $a = \tau b$.

If instead $a_i = 0$ for every $i \geq 1$, then $|a_i| = |b_i|$ implies $b_i = 0$ for every $i \geq 1$, which is exactly alternative (2).

In alternative (1), strong convergence of the head vectors gives

$$\|a_{\mathrm{head}}^{(n)} - \tau b_{\mathrm{head}}^{(n)}\|_2 \leq \|a_{\mathrm{head}}^{(n)} - a\|_2 + \|b_{\mathrm{head}}^{(n)} - b\|_2 \longrightarrow 0.$$

□

Remark 17. *The linear probe in the complex case is exactly the phase analogue of the sign probe in the real case: one replaces* $\mathrm{sign}(\xi)$ *by* $\xi/|\xi|$. *The quadratic probe is unchanged, since* $|\xi|^2 - 1$ *is real.*

Lemma 18 (Limit law for the head-tail decomposition)**.** *With the notation above, set*

$$X_n^{\mathrm{head}} := a_0^{(n)}\eta + \sum_{1\le i\le m_n} a_i^{(n)}\xi_{n,i}, \qquad X_n^{\mathrm{tail}} := \sum_{i>m_n} a_i^{(n)}\xi_{n,i},$$

and define $Y_n^{\mathrm{head}}, Y_n^{\mathrm{tail}}$ *analogously. After passing to a further subsequence, the quadruple*

$$\left(X_n^{\mathrm{head}}, Y_n^{\mathrm{head}}, X_n^{\mathrm{tail}}, Y_n^{\mathrm{tail}}\right)$$

converges in distribution to

$$(H_X, H_Y, R_X, R_Y) \in \mathbb{C}^4,$$

where:

(1) (H_X, H_Y) *is independent of* (R_X, R_Y)*;*
(2)
$$|H_X + R_X| = |H_Y + R_Y| \qquad \textit{almost surely;}$$
(3) *either there exists* $\tau \in \mathbb{T}$ *such that*
$$H_X = \tau H_Y \qquad \textit{almost surely,}$$
or else
$$H_X = a_0\eta, \qquad H_Y = b_0\eta \qquad \textit{almost surely.}$$

Proof. The tightness and independence statements are the same as in the real proof. The last alternative is just Proposition 16 transported to random variables: in alternative (1) of Proposition 16, strong L^2 convergence of the head coefficient vectors gives

$$X_n^{\mathrm{head}} - \tau Y_n^{\mathrm{head}} \longrightarrow 0$$

in L^2, hence $H_X = \tau H_Y$ almost surely. In alternative (2),

$$\sum_{1\le i\le m_n} |a_i^{(n)}|^2 \longrightarrow 0, \qquad \sum_{1\le i\le m_n} |b_i^{(n)}|^2 \longrightarrow 0,$$

so

$$X_n^{\mathrm{head}} - a_0\eta \longrightarrow 0, \qquad Y_n^{\mathrm{head}} - b_0\eta \longrightarrow 0$$

in L^2. □

Proposition 19 (Compactness reduction to an exact head-tail identity)**.** *Assume that Theorem 7 fails, and let the bad sequence be given by Proposition 10. Then, after passing to a subsequence, there exist:*

(1) *an integer sequence* $m_n \to \infty$ *and reordered coefficient vectors* $a^{(n)}, b^{(n)}$ *as in Lemma 11;*
(2) *complex random variables* (H_X, H_Y, R_X, R_Y) *such that* (H_X, H_Y) *is independent of* (R_X, R_Y)*;*
(3) *limiting head coefficients* $a = (a_0, a_1, \dots)$ *and* $b = (b_0, b_1, \dots)$ *satisfying the alternative of Proposition 16;*
(4) *the exact modulus identity*
$$|H_X + R_X| = |H_Y + R_Y| \qquad \textit{almost surely.}$$

Moreover, either there exists $\tau \in \mathbb{T}$ *such that*

$$H_X = \tau H_Y \qquad \textit{almost surely,}$$

or else

$$H_X = a_0\eta, \qquad H_Y = b_0\eta \qquad \textit{almost surely.}$$

Proof. This is exactly the content of Lemma 11, Proposition 16, and Lemma 18. □

Lemma 20 (Lower L^1-L^2 comparability for independent real sums)**.** *Let $\zeta_1, \dots, \zeta_N$ be independent centered real-valued random variables and assume that*

$$\|\zeta_i\|_{L^1} \geq \delta \|\zeta_i\|_{L^2} \qquad (1 \leq i \leq N)$$

for some $\delta > 0$. Then

$$\Big\|\sum_{i=1}^N \zeta_i\Big\|_{L^1} \geq \frac{\delta}{2\sqrt{2}}\Big(\sum_{i=1}^N \|\zeta_i\|_{L^2}^2\Big)^{1/2}.$$

Proof. If $\sum_i \|\zeta_i\|_{L^2}^2 = 0$, there is nothing to prove. By homogeneity, we may therefore assume

$$\sum_{i=1}^N \|\zeta_i\|_{L^2}^2 = 1.$$

Let ζ_i' be an independent copy of ζ_i, and set

$$S := \sum_{i=1}^N \zeta_i, \qquad S' := \sum_{i=1}^N \zeta_i'.$$

Then

$$2\|S\|_{L^1} \geq \|S - S'\|_{L^1} = \Big\|\sum_{i=1}^N (\zeta_i - \zeta_i')\Big\|_{L^1}.$$

The differences $\zeta_i - \zeta_i'$ are independent and symmetric. Conditioning on their absolute values and applying the Khintchine inequality gives

$$\Big\|\sum_{i=1}^N (\zeta_i - \zeta_i')\Big\|_{L^1} \geq \frac{1}{\sqrt{2}}\mathbb{E}\Big[\Big(\sum_{i=1}^N (\zeta_i - \zeta_i')^2\Big)^{1/2}\Big].$$

Since

$$\sum_{i=1}^N c_i^2 |\widetilde{\zeta}_i - \widetilde{\zeta}_i'| \leq \Big(\sum_{i=1}^N c_i^2 (\widetilde{\zeta}_i - \widetilde{\zeta}_i')^2\Big)^{1/2},$$

where we write

$$c_i := \|\zeta_i\|_{L^2}, \qquad \widetilde{\zeta}_i := \begin{cases} \zeta_i / c_i, & c_i \neq 0, \\ 0, & c_i = 0, \end{cases}$$

so that $\sum_i c_i^2 = 1$ and $\|\widetilde{\zeta}_i\|_{L^2} = 1$ whenever $c_i \neq 0$. Applying the previous argument to the sum $\sum_i c_i \widetilde{\zeta}_i$ gives

$$\|S\|_{L^1} \geq \frac{1}{2\sqrt{2}} \sum_{i=1}^N c_i^2 \|\widetilde{\zeta}_i - \widetilde{\zeta}_i'\|_{L^1}.$$

Finally,

$$\|\widetilde{\zeta}_i - \widetilde{\zeta}_i'\|_{L^1} \geq \Big\|\mathbb{E}[\widetilde{\zeta}_i - \widetilde{\zeta}_i' \mid \widetilde{\zeta}_i]\Big\|_{L^1} = \|\widetilde{\zeta}_i\|_{L^1} \geq \delta,$$

because $\widetilde{\zeta}_i'$ is centered and $\|\widetilde{\zeta}_i\|_{L^1} \geq \delta \|\widetilde{\zeta}_i\|_{L^2} = \delta$. Hence

$$\Big\|\sum_{i=1}^N \zeta_i\Big\|_{L^1} \geq \frac{\delta}{2\sqrt{2}}.$$

Undoing the normalization yields the stated inequality. □

Lemma 21 (Infinitely divisible laws supported on a generalized circle are line-supported)**.** *Let μ be an infinitely divisible probability measure on $\mathbb{C}$. If $\operatorname{supp}\mu$ is contained in a generalized circle, then $\operatorname{supp}\mu$ is contained in an affine line.*

Proof. If $\operatorname{supp}\mu$ is already contained in an affine line, there is nothing to prove. It remains to exclude the case in which $\operatorname{supp}\mu$ is contained in a Euclidean circle Γ.

Because μ is infinitely divisible, it is in particular 2-divisible: there exists a probability measure ν such that $\mu = \nu * \nu$. Let $A := \operatorname{supp}\nu$. Then

$$\operatorname{supp}\mu = \overline{A + A}.$$

Assume for contradiction that A contains two distinct points $x \neq y$. Then

$$2x,\ x+y,\ 2y \in A + A \subset \operatorname{supp}\mu \subset \Gamma.$$

But $x + y$ is the midpoint of the chord joining $2x$ and $2y$. The midpoint of a nontrivial chord of a Euclidean circle lies strictly inside the enclosed disk, and therefore cannot belong to the circle itself. This contradiction shows that A is a singleton. Hence $\mu = \nu * \nu$ is a point mass, and in particular its support is contained in an affine line. □

Proposition 22 (Generalized-circle tail rigidity)**.** *Let $(\xi_{n,i})_{n\geq 1,\ i\geq 1}$ be a triangular array of independent centered complex-valued random variables satisfying*

$$\mathbb{E}[|\xi_{n,i}|^2] = 1, \qquad \operatorname{gap}_{\mathrm{gc}}(\xi_{n,i}) \geq \kappa,$$

and the uniform finite-support L^1-L^2 bounds

$$A\|u\|_2 \leq \Big\|\sum_{i\geq 1} u_i \xi_{n,i}\Big\|_{L^1} \leq B\|u\|_2.$$

Let $c^{(n)} \in \ell^2(\mathbb{N})$ satisfy

$$\sup_n \|c^{(n)}\|_2 < \infty, \qquad \max_i |c_i^{(n)}| \longrightarrow 0,$$

and define

$$Z_n := \sum_{i\geq 1} c_i^{(n)} \xi_{n,i}.$$

If Z_n converges in distribution to a random variable Z whose law is supported on a generalized circle, then

$$\|c^{(n)}\|_2 \longrightarrow 0.$$

Proof. Assume for contradiction that $\|c^{(n)}\|_2 \not\to 0$. After passing to a subsequence, we may assume that

$$\delta \leq \|c^{(n)}\|_2 \leq M \qquad \text{for every } n$$

for some constants $\delta > 0$ and $M < \infty$.

Since

$$\mathbb{E}[|Z_n|^2] = \sum_{i\geq 1} |c_i^{(n)}|^2 = \|c^{(n)}\|_2^2 \leq M^2,$$

the family $(Z_n)_n$ is bounded in L^2. In particular, the first moments are uniformly integrable. Because every Z_n is centered, passing to the weak limit gives

$$\mathbb{E}[Z] = 0.$$

The array of summands

$$Z_{n,i} := c_i^{(n)} \xi_{n,i}$$

is infinitesimal: for every $\varepsilon > 0$,

$$\sup_i \mathbb{P}\big(|Z_{n,i}| > \varepsilon\big) \leq \frac{1}{\varepsilon^2} \sup_i |c_i^{(n)}|^2 \longrightarrow 0.$$

By the classical theorem on weak limits of row-wise independent infinitesimal triangular arrays, the law of Z is infinitely divisible. Since by hypothesis it is also supported on a generalized circle,

Lemma 21 shows that $\operatorname{supp}(Z)$ is contained in an affine line. Because $\mathbb{E}[Z] = 0$, that line passes through the origin. Hence there exists $\lambda \in \mathbb{T}$ such that

$$\lambda Z \in \mathbb{R} \qquad \text{almost surely.}$$

Define

$$Y_n := \Im(\lambda Z_n) = \sum_{i\geq 1} Y_{n,i}, \qquad Y_{n,i} := \Im(\lambda c_i^{(n)} \xi_{n,i}).$$

Then each $Y_{n,i}$ is a centered real random variable, the family $(Y_{n,i})_i$ is independent for each fixed n, and

$$Y_n \Rightarrow 0$$

in distribution.

We claim that each nonzero summand $Y_{n,i}$ satisfies a uniform lower L^1-L^2 comparability estimate. If $c_i^{(n)} \neq 0$, write

$$\omega_{n,i} := \lambda \frac{c_i^{(n)}}{|c_i^{(n)}|} \in \mathbb{T}.$$

Then

$$Y_{n,i} = |c_i^{(n)}|\, \Im(\omega_{n,i}\xi_{n,i}).$$

For every $\omega \in \mathbb{T}$, choose

$$A = 0, \qquad C = 0, \qquad B = -\frac{i\overline{\omega}}{\sqrt{2}}.$$

Then

$$A^2 + 2|B|^2 + C^2 = 1$$

and

$$A|\xi|^2 + 2\Re(B\xi) + C = \sqrt{2}\,\Im(\omega\xi).$$

Therefore

$$\kappa \leq \operatorname{gap}_{\mathrm{gc}}(\xi_{n,i}) \leq \sqrt{2}\,\|\Im(\omega_{n,i}\xi_{n,i})\|_{L^1} \leq \sqrt{2}\,\|\Im(\omega_{n,i}\xi_{n,i})\|_{L^2}.$$

Multiplying by $|c_i^{(n)}|$ gives

$$\|Y_{n,i}\|_{L^1} \geq \frac{\kappa}{\sqrt{2}}|c_i^{(n)}|, \qquad \|Y_{n,i}\|_{L^2} \geq \frac{\kappa}{\sqrt{2}}|c_i^{(n)}|.$$

Since also $\|Y_{n,i}\|_{L^2} \leq |c_i^{(n)}|$, we get

$$\|Y_{n,i}\|_{L^1} \geq \frac{\kappa}{\sqrt{2}}\|Y_{n,i}\|_{L^2}.$$

Applying Lemma 20 row by row, we obtain

$$\|Y_n\|_{L^1} \geq \frac{\kappa}{4}\Big(\sum_{i\geq 1}\|Y_{n,i}\|_{L^2}^2\Big)^{1/2} \geq \frac{\kappa^2}{4\sqrt{2}}\|c^{(n)}\|_2 \geq \frac{\kappa^2\delta}{4\sqrt{2}}.$$

On the other hand, $Y_n \Rightarrow 0$ and

$$\sup_n \mathbb{E}[|Y_n|^2] \leq \sup_n \mathbb{E}[|Z_n|^2] \leq M^2,$$

so the family $(|Y_n|)_n$ is uniformly integrable. Hence

$$\|Y_n\|_{L^1} \longrightarrow 0,$$

which contradicts the uniform positive lower bound above. □

Remark 23. *Proposition 22 is the complex replacement for the one-dimensional tail rigidity used in the real proof. Its proof separates the generalized-circle obstruction into two parts: infinite divisibility rules out the curved case, and the remaining line case reduces to the real L^1-L^2 argument after taking imaginary parts in the correct direction.*

5. Tail support lemmas

Lemma 24 (Geometric reduction when the head is nontrivial)**.** *Let H be a complex-valued random variable, let (R_X, R_Y) be a complex-valued random vector independent of H, and let $\tau \in \mathbb{T}$. Assume that*

$$|H + R_X| = |\tau H + R_Y| \qquad \text{almost surely.}$$

If $\operatorname{supp}(H)$ *contains at least two distinct points, then the law of*

$$W := R_Y - \tau R_X$$

is supported on an affine line.

Proof. By independence and Lemmas 5 and 6, the identity

$$|H + R_X| = |\tau H + R_Y|$$

holds for every

$$h \in \operatorname{supp}(H), \qquad (x, y) \in \operatorname{supp}(R_X, R_Y).$$

Choose two distinct points $h_1, h_2 \in \operatorname{supp}(H)$. Fix $(x, y) \in \operatorname{supp}(R_X, R_Y)$ and write

$$w := y - \tau x.$$

For $j = 1, 2$ one has

$$|h_j + x| = |\tau(h_j + x) + w|.$$

Squaring and cancelling $|h_j + x|^2$ gives

$$|w|^2 + 2\Re\big(\overline{\tau(h_j + x)}\, w\big) = 0 \qquad (j = 1, 2).$$

Subtracting the two identities yields

$$\Re\big(\overline{\tau(h_1 - h_2)}\, w\big) = 0.$$

Thus every point of the image of $\operatorname{supp}(R_X, R_Y)$ under the continuous map

$$(x, y) \mapsto y - \tau x$$

lies in the affine line

$$L := \{w \in \mathbb{C} : \ \Re(\overline{\tau(h_1 - h_2)}\, w) = 0\}.$$

Since L is closed, the support of the pushforward law of W is also contained in L. □

Remark 25. *This section collects the support lemmas used in the final compactness argument. The nontrivial-head case is handled by a direct geometric elimination of the head variable. The degenerate case $H_X \equiv H_Y \equiv 0$ is treated by classifying additive subgroups of the cone*

$$\Gamma := \{(u, v) \in \mathbb{C}^2 : \ |u| = |v|\}.$$

Lemma 26 (Two distinct unimodular slopes force real-line degeneracy)**.** *Let $\tau_1, \tau_2 \in \mathbb{T}$ with $\tau_1 \neq \tau_2$, and for $j = 1, 2$ let*

$$L_{\tau_j} := \{(z, \tau_j z) : z \in \mathbb{C}\} \subset \mathbb{C}^2.$$

Let $S \subset (L_{\tau_1} \cup L_{\tau_2}) \cap \{(u, v) \in \mathbb{C}^2 : \ |u| = |v|\}$ and assume that

$$S + S \subset \{(u, v) \in \mathbb{C}^2 : \ |u| = |v|\}.$$

For $j = 1, 2$, define

$$K_j := \{z \in \mathbb{C} : \ (z, \tau_j z) \in S\}.$$

Then the following hold.

(1) *For every* $z_1 \in K_1$ *and* $z_2 \in K_2$,

$$\Re\big((1-\tau_1\overline{\tau_2})z_1\overline{z_2}\big) = 0.$$

(2) *If* K_2 *contains a nonzero point, then* K_1 *is contained in a real line through the origin.*
(3) *If* K_1 *and* K_2 *both contain nonzero points, then both sets are contained in real lines through the origin.*

Proof. Fix $z_1 \in K_1$ and $z_2 \in K_2$. Then

$$x := (z_1, \tau_1 z_1) \in S, \qquad y := (z_2, \tau_2 z_2) \in S.$$

By hypothesis, $x + y \in S + S$ belongs to the cone, so

$$|z_1 + z_2| = |\tau_1 z_1 + \tau_2 z_2|.$$

Squaring and using $|\tau_1| = |\tau_2| = 1$, we obtain

$$|z_1|^2 + |z_2|^2 + 2\Re(z_1\overline{z_2}) = |z_1|^2 + |z_2|^2 + 2\Re(\tau_1\overline{\tau_2}\, z_1\overline{z_2}),$$

hence

$$\Re\big((1-\tau_1\overline{\tau_2})z_1\overline{z_2}\big) = 0.$$

This proves (1).

Assume now that $z_2 \in K_2$ is nonzero. Since $\tau_1 \neq \tau_2$, the complex number

$$\lambda := (1-\tau_1\overline{\tau_2})\overline{z_2}$$

is nonzero. By (1), every $z_1 \in K_1$ satisfies

$$\Re(\lambda z_1) = 0,$$

which describes a real line through the origin in $\mathbb{C} \cong \mathbb{R}^2$. This proves (2). Interchanging the roles of the two slopes gives (3). □

Lemma 27 (Covariance range of a non-line-supported rank-one complex vector)**.** *Let* ξ *be a centered complex-valued random variable with*

$$\mathbb{E}[|\xi|^2] < \infty,$$

and let $(a, b) \in \mathbb{C}^2 \setminus \{(0,0)\}$. *Define*

$$V := (a\xi, b\xi) \in \mathbb{C}^2.$$

Assume that ξ *is not supported on a real line in* $\mathbb{C}$. *Then the range of the real covariance operator of* V *is exactly the real plane underlying the complex line*

$$L := \{(az, bz) : z \in \mathbb{C}\}.$$

Proof. Since V is supported on L, the covariance range is contained in the underlying real plane $L_{\mathbb{R}}$.

Conversely, let $\ell : \mathbb{C}^2 \to \mathbb{R}$ be a real linear functional in the kernel of the covariance operator. Because V is centered, this means

$$\mathbb{E}[\ell(V)^2] = 0,$$

hence $\ell(V) = 0$ almost surely. By continuity, ℓ vanishes on $\mathrm{supp}(V)$. Since ξ is not supported on a real line, the real span of $\mathrm{supp}(\xi)$ is all of $\mathbb{C}$, and therefore the real span of $\mathrm{supp}(V)$ is the full plane $L_{\mathbb{R}}$. Thus ℓ vanishes on $L_{\mathbb{R}}$, so the annihilator of the covariance range equals the annihilator of $L_{\mathbb{R}}$. Hence the covariance range is exactly $L_{\mathbb{R}}$. □

Lemma 28 (Gaussian part in the cone lies in one complex line)**.** *Let μ be an infinitely divisible probability measure on $\mathbb{C}^2$ supported on the cone*

$$\Gamma := \{(u, v) \in \mathbb{C}^2 : \ |u| = |v|\}.$$

Assume that μ arises as a weak limit of sums of independent infinitesimal vectors

$$V_{n,i} := (a_{n,i}\xi_{n,i},\, b_{n,i}\xi_{n,i}),$$

where each $\xi_{n,i}$ is centered, square-integrable, and satisfies

$$\mathrm{gap}_{\mathrm{gc}}(\xi_{n,i}) > 0.$$

Then the Gaussian part of μ is supported on a single complex line

$$L_\tau := \{(z, \tau z) : z \in \mathbb{C}\} \qquad \textit{for some } \tau \in \mathbb{T}.$$

Proof. Let G denote the Gaussian part of μ, and let $E \subset \mathbb{C}^2 \cong \mathbb{R}^4$ be its support subspace. Since μ is supported on Γ, the same is true of G, so

$$E \subset \Gamma.$$

The Gaussian covariance of the limit is obtained from the accumulation of the row-wise covariance operators of the infinitesimal summands. By Lemma 27, every nonzero covariance contribution is the real plane underlying a complex line

$$L_{n,i} := \{(a_{n,i}z, b_{n,i}z) : z \in \mathbb{C}\},$$

because $\mathrm{gap}_{\mathrm{gc}}(\xi_{n,i}) > 0$ rules out support on a real line.

Hence the Gaussian support subspace E is generated, as a real vector space, by such planes $L_{n,i,\mathbb{R}}$. Since $E \subset \Gamma$, every contributing plane is itself contained in Γ. Therefore, whenever $(a_{n,i}, b_{n,i}) \neq (0, 0)$ and the corresponding covariance contribution is nonzero, one must have

$$|a_{n,i}| = |b_{n,i}|,$$

so the associated complex line is of the form

$$L_{\tau_{n,i}} := \{(z, \tau_{n,i}z) : z \in \mathbb{C}\} \qquad \text{with } \tau_{n,i} \in \mathbb{T}.$$

Assume for contradiction that two distinct slopes $\tau_1 \neq \tau_2$ occur among the planes contributing to E. Then E contains both complex lines L_{τ_1} and L_{τ_2}. Since E is a real vector subspace, it is closed under addition, so

$$(L_{\tau_1} \cup L_{\tau_2}) + (L_{\tau_1} \cup L_{\tau_2}) \subset E \subset \Gamma.$$

Applying Lemma 26 with

$$S := L_{\tau_1} \cup L_{\tau_2}$$

shows that the scalar set on each slope must be contained in a real line through the origin. This is impossible because each contributing set is the full complex plane of scalars. Hence only one slope can occur.

Therefore all covariance contributions lie in one complex line L_τ, and so their sum, namely the Gaussian covariance of μ, has range contained in the underlying real plane $L_{\tau,\mathbb{R}}$. Thus G is supported on L_τ. □

Lemma 29 (Support of the Lévy measure inherits cone additivity)**.** *Let μ be an infinitely divisible probability measure on $\mathbb{C}^2$ with Lévy measure ν, and assume that*

$$\mathrm{supp}(\mu) \subset \Gamma := \{(u, v) \in \mathbb{C}^2 : \ |u| = |v|\}.$$

Then:

(1) $\mathrm{supp}(\nu) \subset \Gamma \cup \{0\}$*;*
(2) *if $x, y \in \mathrm{supp}(\nu) \setminus \{0\}$, then*

$$x + y \in \Gamma.$$

Proof. Fix $x \in \operatorname{supp}(\nu) \setminus \{0\}$. Choose an open neighborhood U of x whose closure is bounded away from the origin. Then $0 < \nu(U) < \infty$. Write

$$\nu = \nu|_U + \nu|_{U^c}$$

and let μ_U and μ_{U^c} be the corresponding infinitely divisible laws, so that

$$\mu = \mu_U * \mu_{U^c}.$$

Since $\nu(U) < \infty$, the law μ_U is a compound Poisson law. In particular, with positive probability it has exactly one jump in U and no other jumps, so every point of U belongs to $\operatorname{supp}(\mu_U)$. As convolution supports add, this implies

$$U \subset \operatorname{supp}(\mu_U) \subset \operatorname{supp}(\mu) - \operatorname{supp}(\mu_{U^c}).$$

Taking closures and using $\operatorname{supp}(\mu) \subset \Gamma$ yields $x \in \Gamma$. This proves (1).

Now let $x, y \in \operatorname{supp}(\nu) \setminus \{0\}$. Choose disjoint open neighborhoods U, V of x, y whose closures are bounded away from 0. Then $0 < \nu(U), \nu(V) < \infty$. Let $\mu_{U,V}$ denote the compound Poisson law with Lévy measure $\nu|_{U \cup V}$. With positive probability, a sample from $\mu_{U,V}$ has exactly one jump in U, exactly one jump in V, and no other jumps in $U \cup V$. Hence

$$U + V \subset \operatorname{supp}(\mu_{U,V}).$$

As before, since $\mu = \mu_{U,V} * \mu_{(U \cup V)^c}$ and $\operatorname{supp}(\mu) \subset \Gamma$, it follows that every point of $U + V$ belongs to Γ. Letting the neighborhoods shrink to x and y gives

$$x + y \in \Gamma.$$

□

Corollary 30 (Two slopes in the Lévy measure force real-line degeneracy)**.** *Let μ be an infinitely divisible probability measure on $\mathbb{C}^2$ supported on*

$$\Gamma := \{(u, v) \in \mathbb{C}^2 : \ |u| = |v|\},$$

and let ν be its Lévy measure. Suppose ν charges two distinct unimodular slope lines

$$L_{\tau_1} = \{(z, \tau_1 z) : z \in \mathbb{C}\}, \qquad L_{\tau_2} = \{(z, \tau_2 z) : z \in \mathbb{C}\}, \qquad \tau_1 \neq \tau_2.$$

Define

$$K_j := \{z \in \mathbb{C} : \ (z, \tau_j z) \in \operatorname{supp}(\nu)\} \qquad (j = 1, 2).$$

Then every nonzero point of K_1 forces K_2 into a real line through the origin, and conversely.

Proof. Apply Lemma 29 with

$$S := \operatorname{supp}(\nu) \setminus \{0\}.$$

Then $S \subset \Gamma$ and $S + S \subset \Gamma$. Lemma 26 therefore applies. □

Remark 31. *Corollary 30 is the jump-part analogue of Lemma 26. It shows that the only possible way for the Lévy measure to charge two distinct unimodular slopes is through a real-line degeneracy of the scalar jump sets on those slopes.*

Lemma 32 (Cone-supported infinitely divisible laws live on additive subgroups)**.** *Let μ be an infinitely divisible probability measure on*

$$\Gamma := \{(u, v) \in \mathbb{C}^2 : \ |u| = |v|\}.$$

Then there exists an additive subgroup $G \subset \mathbb{C}^2$ such that

$$\operatorname{supp}(\mu) \subset \overline{G} \subset \Gamma.$$

Proof. Since μ is infinitely divisible, there exists a probability measure ν such that

$$\mu = \nu * \nu.$$

Let

$$A := \operatorname{supp}(\nu).$$

By the support formula for convolutions on $\mathbb{R}^4 \cong \mathbb{C}^2$,

$$\operatorname{supp}(\mu) = \operatorname{supp}(\nu * \nu) = \overline{A + A}.$$

Since $\operatorname{supp}(\mu) \subset \Gamma$ and Γ is closed, it follows that

$$A + A \subset \Gamma.$$

Define the real quadratic form

$$Q(u, v) := |u|^2 - |v|^2$$

and the associated symmetric real bilinear form

$$B\big((u, v), (u', v')\big) := \Re(u\overline{u'} - v\overline{v'}).$$

Then

$$Q(x + y) = Q(x) + Q(y) + 2B(x, y).$$

If $x \in A$, then $2x = x + x \in A + A \subset \Gamma$, so

$$0 = Q(2x) = 4Q(x),$$

hence $x \in \Gamma$. Thus

$$A \subset \Gamma.$$

Now if $x, y \in A$, then $x + y \in A + A \subset \Gamma$, and therefore

$$0 = Q(x + y) = Q(x) + Q(y) + 2B(x, y) = 2B(x, y),$$

because $Q(x) = Q(y) = 0$. Hence

$$B(x, y) = 0 \qquad (x, y \in A).$$

Let G be the additive subgroup generated by A. Thus every $g \in G$ has the form

$$g = \sum_{j=1}^{m} n_j a_j, \qquad n_j \in \mathbb{Z},\ a_j \in A.$$

Using bilinearity and the pairwise orthogonality just proved,

$$Q(g) = \sum_{j=1}^{m} n_j^2 Q(a_j) + 2 \sum_{1 \le i < j \le m} n_i n_j B(a_i, a_j) = 0.$$

So

$$G \subset \Gamma.$$

Finally, since $A + A \subset G$,

$$\operatorname{supp}(\mu) = \overline{A + A} \subset \overline{G} \subset \Gamma.$$

□

Lemma 33 (Additive subgroups of the cone)**.** *Let $G \subset \Gamma$ be an additive subgroup, where*

$$\Gamma := \{(u, v) \in \mathbb{C}^2 :\ |u| = |v|\}.$$

Then there exists $\lambda \in \mathbb{T}$ such that

$$G \subset P_\lambda^+ \qquad \text{or} \qquad G \subset P_\lambda^-,$$

where

$$P_\lambda^+ := \{(z, \lambda z) : z \in \mathbb{C}\}, \qquad P_\lambda^- := \{(z, \lambda \overline{z}) : z \in \mathbb{C}\}.$$

Proof. Let

$$H := \operatorname{span}_{\mathbb{R}}(G) \subset \mathbb{C}^2 \cong \mathbb{R}^4.$$

Using the same forms

$$Q(u,v) := |u|^2 - |v|^2, \qquad B\big((u,v),(u',v')\big) := \Re(u\overline{u'} - v\overline{v'}),$$

we have, for every $x, y \in G$,

$$0 = Q(x+y) - Q(x) - Q(y) = 2B(x,y),$$

because $x, y, x+y \in \Gamma$. Thus

$$B(x,y) = 0 \qquad (x, y \in G).$$

By bilinearity, B vanishes on $H \times H$, so H is a totally isotropic real subspace for the form B. Since B has signature $(2,2)$ on $\mathbb{R}^4$, one has

$$\dim_{\mathbb{R}} H \le 2.$$

If $\dim_{\mathbb{R}} H \le 1$, then every nonzero vector of H belongs to Γ, and the line H is contained in both some P_λ^+ and some P_λ^-. So the conclusion is immediate.

Assume now that $\dim_{\mathbb{R}} H = 2$. Consider the projection

$$\pi_1 : H \to \mathbb{C}, \qquad \pi_1(u,v) := u.$$

If $(0,v) \in H$, then

$$|v| = |0| = 0,$$

so $(0,v) = 0$. Thus π_1 is injective. Since both H and $\mathbb{C}$ are 2-dimensional over $\mathbb{R}$, the map π_1 is a real-linear isomorphism. Hence H is the graph of a real-linear map $T : \mathbb{C} \to \mathbb{C}$:

$$H = \{(z, Tz) : z \in \mathbb{C}\}.$$

Because $H \subset \Gamma$, one has

$$|Tz| = |z| \qquad (z \in \mathbb{C}),$$

so T is a real-linear isometry of $\mathbb{C} \cong \mathbb{R}^2$. Therefore either

$$Tz = \lambda z \qquad \text{for all } z \in \mathbb{C},$$

or

$$Tz = \lambda \overline{z} \qquad \text{for all } z \in \mathbb{C},$$

for some $\lambda \in \mathbb{T}$. Thus

$$H = P_\lambda^+ \qquad \text{or} \qquad H = P_\lambda^-.$$

Since $G \subset H$, the proof is complete. □

Remark 34. *Lemmas 32 and 33 give an alternative structural description of cone-supported infinitely divisible laws: their support is contained in the closure of an additive subgroup lying either in a complex line P_λ^+ or in a conjugate plane P_λ^-. This is consistent with the Gaussian obstruction $(Z, \overline{Z})$ and explains why the rank-one structure of the array is needed to rule out the conjugate-plane alternative.*

Lemma 35 (Diffuse tails force a nontrivial Gaussian part)**.** *Let*

$$V_{n,i} := (a_i^{(n)} \xi_{n,i},\, b_i^{(n)} \xi_{n,i}) \in \mathbb{C}^2 \cong \mathbb{R}^4$$

for $i > m_n$, where the coordinates $\xi_{n,i}$ are independent and satisfy

$$\mathbb{E}[\xi_{n,i}] = 0, \qquad \mathbb{E}[|\xi_{n,i}|^2] = 1, \qquad \mathbb{E}[|\xi_{n,i}|] \ge A > 0.$$

Define

$$\alpha_{n,i} := \big(|a_i^{(n)}|^2 + |b_i^{(n)}|^2\big)^{1/2}.$$

Assume that

$$\max_{i>m_n} \alpha_{n,i} \longrightarrow 0, \qquad \sum_{i>m_n} \alpha_{n,i}^2 \longrightarrow \sigma^2 > 0,$$

and that the row sums

$$\sum_{i>m_n} V_{n,i}$$

converge in distribution to an infinitely divisible law μ on $\mathbb{C}^2$. Then the Gaussian part of μ is nontrivial.

Proof. Let

$$\beta_n := \max_{i>m_n} \alpha_{n,i}, \qquad r_n := \beta_n^{1/2}.$$

Then

$$r_n \downarrow 0, \qquad \frac{r_n}{\beta_n} \longrightarrow \infty.$$

For each $i > m_n$ with $\alpha_{n,i} \neq 0$, set

$$K_{n,i} := \frac{r_n}{\alpha_{n,i}} \geq \frac{r_n}{\beta_n} = \beta_n^{-1/2},$$

so $K_{n,i} \to \infty$ uniformly in $i > m_n$. Define the centered truncation

$$\widetilde{\xi}_{n,i} := \xi_{n,i}\mathbf{1}_{\{|\xi_{n,i}|\leq K_{n,i}\}} - \mathbb{E}\Big[\xi_{n,i}\mathbf{1}_{\{|\xi_{n,i}|\leq K_{n,i}\}}\Big],$$

and set

$$\widetilde{V}_{n,i} := (a_i^{(n)}\widetilde{\xi}_{n,i},\, b_i^{(n)}\widetilde{\xi}_{n,i}).$$

If $\alpha_{n,i} = 0$, simply set $\widetilde{\xi}_{n,i} = 0$ and $\widetilde{V}_{n,i} = 0$.

For every $K > 0$ and every centered ξ with $\mathbb{E}|\xi|^2 = 1$ and $\mathbb{E}|\xi| \geq A$, one has

$$\mathbb{E}|\xi| \leq \big(\mathbb{E}\big[|\xi|^2\mathbf{1}_{\{|\xi|\leq K\}}\big]\big)^{1/2} + \mathbb{E}\big[|\xi|\mathbf{1}_{\{|\xi|>K\}}\big] \leq \big(\mathbb{E}\big[|\xi|^2\mathbf{1}_{\{|\xi|\leq K\}}\big]\big)^{1/2} + \frac{1}{K},$$

and therefore

$$\mathbb{E}\big[|\xi|^2\mathbf{1}_{\{|\xi|\leq K\}}\big] \geq \left(A - \frac{1}{K}\right)^2.$$

Also, since $\mathbb{E}[\xi] = 0$,

$$\big|\mathbb{E}\big[\xi\mathbf{1}_{\{|\xi|\leq K\}}\big]\big| = \big|\mathbb{E}\big[\xi\mathbf{1}_{\{|\xi|>K\}}\big]\big| \leq \mathbb{E}\big[|\xi|\mathbf{1}_{\{|\xi|>K\}}\big] \leq \frac{1}{K}.$$

Hence

$$\mathrm{Var}\big(\xi\mathbf{1}_{\{|\xi|\leq K\}} - \mathbb{E}\big[\xi\mathbf{1}_{\{|\xi|\leq K\}}\big]\big) \geq \left(A - \frac{1}{K}\right)^2 - \frac{1}{K^2}.$$

Choose K_0 large enough that

$$c_0 := \left(A - \frac{1}{K_0}\right)^2 - \frac{1}{K_0^2} > 0.$$

Since $K_{n,i} \to \infty$ uniformly, for all sufficiently large n one has $K_{n,i} \geq K_0$ for every $i > m_n$. Therefore

$$\mathbb{E}|\widetilde{\xi}_{n,i}|^2 \geq c_0 \qquad (i > m_n).$$

Next,

$$|\widetilde{V}_{n,i}| \leq |V_{n,i}|\mathbf{1}_{\{|V_{n,i}|\leq r_n\}} + \mathbb{E}\Big[|V_{n,i}|\mathbf{1}_{\{|V_{n,i}|\leq r_n\}}\Big] \leq 2r_n,$$

so the centered truncation array is bounded and infinitesimal. Moreover,

$$\mathbb{E}|\widetilde{V}_{n,i}|^2 = \alpha_{n,i}^2\,\mathbb{E}|\widetilde{\xi}_{n,i}|^2 \geq c_0\,\alpha_{n,i}^2$$

for all large n. Summing over $i > m_n$ gives

$$\sum_{i>m_n} \mathbb{E}|\widetilde{V}_{n,i}|^2 \geq c_0 \sum_{i>m_n} \alpha_{n,i}^2 \longrightarrow c_0\sigma^2 > 0.$$

By the standard shrinking-truncation theorem for infinitesimal triangular arrays and infinitely divisible limits, the covariance matrices of these centered truncations converge to the Gaussian covariance matrix of μ; see for instance Parthasarathy [7, Chapter IV, §5, pp. 82–96; §6, pp. 97–101; §7, pp. 102–108]; these are the passages where the accompanying infinitely divisible laws are constructed and the Gaussian component is identified through the covariance of shrinking centered truncations. Since the traces stay bounded away from zero, that Gaussian covariance matrix is nonzero. Thus the Gaussian part of μ is nontrivial. □

Lemma 36 (Degenerate cone case reduces to a line)**.** *Let (R_X, R_Y) be the tail limit given by Lemma 18, and assume*

$$H_X \equiv 0, \qquad H_Y \equiv 0.$$

Then there exists $\tau \in \mathbb{T}$ such that the law of

$$R_Y - \tau R_X$$

is supported on an affine line.

Proof. Let

$$\mu := \mathrm{law}(R_X, R_Y).$$

Since $H_X \equiv H_Y \equiv 0$, Lemma 18 gives

$$|R_X| = |R_Y| \qquad \text{almost surely,}$$

so

$$\mathrm{supp}(\mu) \subset \Gamma := \{(u, v) \in \mathbb{C}^2 : \ |u| = |v|\}.$$

By Lemmas 32 and 33, there exists $\lambda \in \mathbb{T}$ such that

$$\mathrm{supp}(\mu) \subset P_\lambda^+ \qquad \text{or} \qquad \mathrm{supp}(\mu) \subset P_\lambda^-,$$

where

$$P_\lambda^+ := \{(z, \lambda z) : z \in \mathbb{C}\}, \qquad P_\lambda^- := \{(z, \lambda \overline{z}) : z \in \mathbb{C}\}.$$

If $\mathrm{supp}(\mu) \subset P_\lambda^+$, then

$$R_Y = \lambda R_X \qquad \text{almost surely,}$$

so choosing $\tau := \lambda$ gives

$$R_Y - \tau R_X = 0 \qquad \text{almost surely.}$$

This is already stronger than the conclusion.

It remains to treat the case

$$\mathrm{supp}(\mu) \subset P_\lambda^-.$$

Then for every support point of μ there exists $z \in \mathbb{C}$ such that

$$(R_X, R_Y) = (z, \lambda \overline{z}).$$

Choosing $\tau := \lambda$, we obtain

$$R_Y - \tau R_X = \lambda(\overline{z} - z) \in i\lambda\,\mathbb{R}.$$

Thus the law of $R_Y - \tau R_X$ is supported on the real line

$$i\lambda\,\mathbb{R} \subset \mathbb{C}.$$

□

Remark 37. *Lemma 35 is exactly where the uniform lower L^1 bound enters the degenerate-head case. It gives a positive lower bound for the variances of shrinking centered truncations without any appeal to uniform L^2-integrability of the coordinate family.*

Remark 38. *The jump-part lemmas above remain useful as complementary structural information. In particular, Corollary 30 shows that any attempt to prove the theorem using only the Lévy measure would still be forced into a real-line-degenerate two-slope analysis.*

Remark 39 (Historical route via quadratic mass)**.** *Before Lemma 35 was isolated, the degenerate case was naturally led by the observation*

$$\sum_{i>m_n} \big(|a_i^{(n)}|^2 + |b_i^{(n)}|^2\big) \longrightarrow 2.$$

Lemma 35 is the formal version of that heuristic: the shrinking-truncation method does produce a nonzero Gaussian part once the uniform lower L^1 bound is fed into the truncation estimates.

6. Proof of Theorem 7

Proof of Theorem 7. Assume for contradiction that Theorem 7 fails. Let the bad sequence be given by Proposition 10. Apply Proposition 19. Thus, after passing to a subsequence, there exist head and tail limits (H_X, H_Y, R_X, R_Y) with (H_X, H_Y) independent of (R_X, R_Y) and

$$|H_X + R_X| = |H_Y + R_Y| \qquad \text{almost surely.}$$

Case 1: H_X is not almost surely constant. By Proposition 19, there exists $\tau \in \mathbb{T}$ such that

$$H_Y = \tau H_X \qquad \text{almost surely.}$$

Since $\operatorname{supp}(H_X)$ contains at least two distinct points, Lemma 24 shows that the law of

$$W := R_Y - \tau R_X$$

is supported on an affine line, hence on a generalized circle.

Now

$$W_n := Y_n^{\text{tail}} - \tau X_n^{\text{tail}} = \sum_{i>m_n} (b_i^{(n)} - \tau a_i^{(n)})\, \xi_{n,i}$$

converges in distribution to W. The coefficients are diffuse because Lemma 11 gives

$$\max_{i>m_n} |b_i^{(n)} - \tau a_i^{(n)}| \le 2 \max_{i>m_n} \max\big(|a_i^{(n)}|, |b_i^{(n)}|\big) \longrightarrow 0.$$

Therefore Proposition 22 implies

$$\|b_{\text{tail}}^{(n)} - \tau a_{\text{tail}}^{(n)}\|_2 \longrightarrow 0.$$

By Proposition 16,

$$\|b_{\text{head}}^{(n)} - \tau a_{\text{head}}^{(n)}\|_2 \longrightarrow 0.$$

Hence

$$\|b^{(n)} - \tau a^{(n)}\|_2 \longrightarrow 0.$$

This contradicts the orthogonal normalization, since

$$\|b^{(n)} - \tau a^{(n)}\|_2^2 = \|a^{(n)}\|_2^2 + \|b^{(n)}\|_2^2 - 2\Re\big(\tau\langle a^{(n)}, b^{(n)}\rangle\big) = 2.$$

Case 2: H_X is almost surely constant. Since H_X is centered as a weak limit of centered random variables, one has

$$H_X \equiv 0, \qquad H_Y \equiv 0.$$

By Lemma 36, there exists $\tau_0 \in \mathbb{T}$ such that the law of

$$W := R_Y - \tau_0 R_X$$

is supported on an affine line, hence on a generalized circle. Therefore Proposition 22 applied to

$$W_n := Y_n^{\text{tail}} - \tau_0 X_n^{\text{tail}}$$

shows that

$$\|b_{\text{tail}}^{(n)} - \tau_0 a_{\text{tail}}^{(n)}\|_2 \longrightarrow 0.$$

Because $H_X \equiv H_Y \equiv 0$, the head limits vanish, so

$$\|a_{\text{head}}^{(n)}\|_2 \longrightarrow 0, \qquad \|b_{\text{head}}^{(n)}\|_2 \longrightarrow 0.$$

Thus

$$\|b^{(n)} - \tau_0 a^{(n)}\|_2 \longrightarrow 0,$$

again contradicting the orthogonal normalization. □

Remark 40 (Alternative Gaussian route in the degenerate case)**.** *The proof of Lemma 36 does not use the Gaussian part. Nevertheless, the same conclusion also follows from Lemma 35. Indeed, if*

$$|R_X| = |R_Y|$$

and one writes the tail law as a convolution of its Gaussian part and its independent remainder, then the Gaussian part is nontrivial and, by Lemma 28, lies in a single complex line

$$\{(z, \tau z) : z \in \mathbb{C}\}.$$

Using the exact modulus identity and Lemma 24 with the Gaussian first coordinate as the new head again forces $R_Y - \tau R_X$ onto an affine line. Thus the degenerate case admits both a pure support proof and an independent Gaussian-based proof.

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